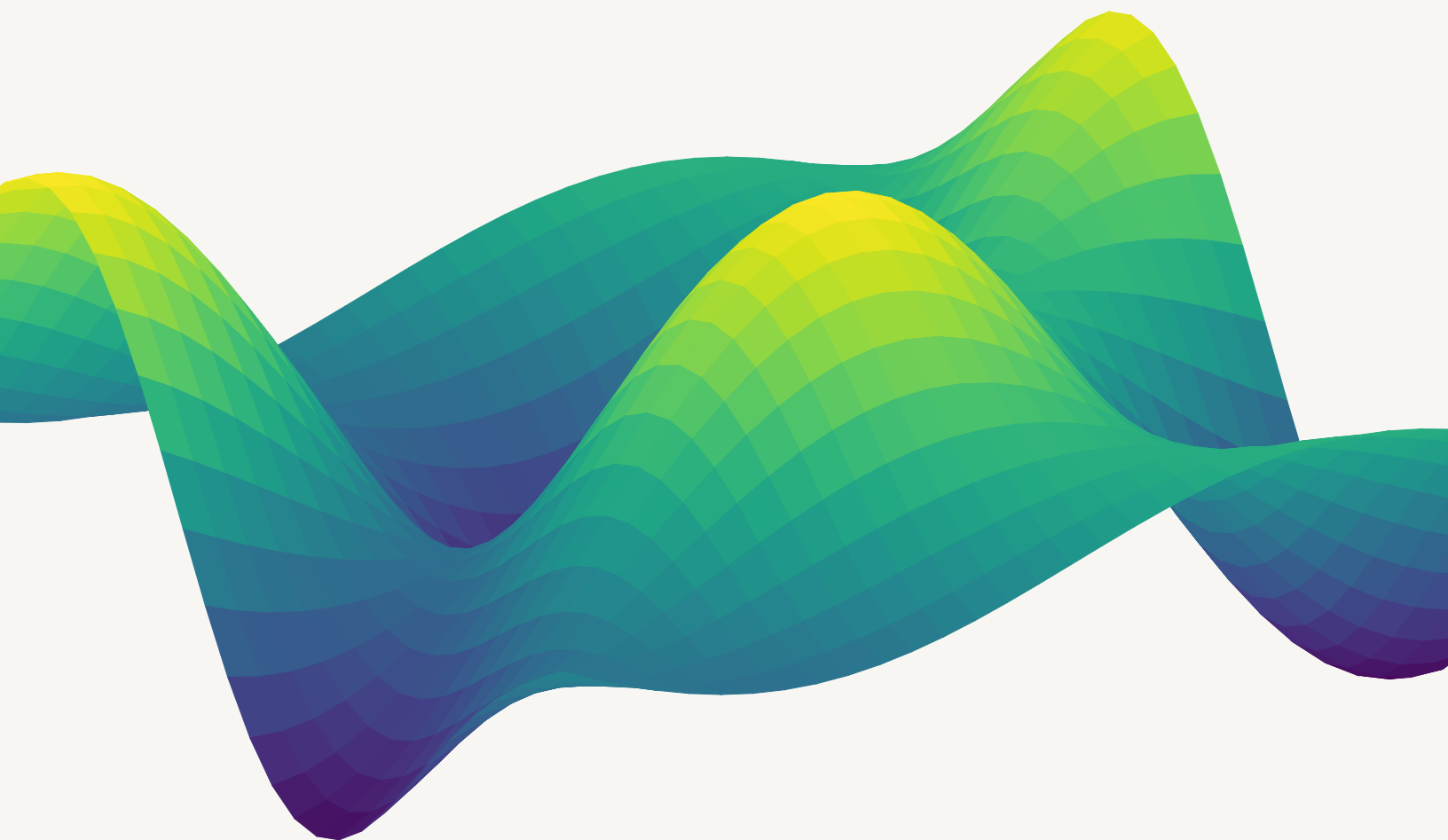


MATHEMATICAL METHODS II

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Vector Calculus ◦ Integral Theorems ◦ PDEs

Boundary Value Problems ◦ Green's Method



Abstract

These notes are a summary of the Mathematical Methods 2 module taught by Anthony Yeates and Christopher Prior in 2025-26. There will be minimal, if any, proofs as these notes focus on the main results of the course for revision purposes. If you do find any errors or typos, feel free to let me know at davidpuertamaths@gmail.com

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Notation

$\mathbf{x}$ – a vector

x – a scalar

x_i – the i^{th} component of $\mathbf{x}$

$:=$ – a definition

$\dot{\mathbf{x}}$ – the time derivative of $\mathbf{x}$

$\subset$ – a subset of

$\mathbb{R}^n$ – the n dimensional vector space of real numbers

$\{\mathbf{e}_i\}_{i=1}^n$ – the standard basis of $\mathbb{R}^n$

$[a, b]$ – the interval $a \leq x \leq b$ for $a, b, x \in \mathbb{R}$

$\oint$ – the integral around a closed curve

$d\mathbf{x}$ – an infinitesimal displacement vector along a curve/surface

∂A – the boundary of a n -dimensional surface A

∇ – the grad differential operator

$\equiv$ – an identity, not an equality

∂_i – the partial derivative in the i^{th} coordinate, that is $\partial_i := \frac{\partial}{\partial x_i}$

$\mathcal{L}[\cdot]$ – a linear operator

$\mathcal{L}^*[\cdot]$ – the adjoint linear operator to the above

BC_i – the i^{th} boundary condition

BC_i^* – the i^{th} adjoint boundary condition

$\mathcal{G}(x, \xi)$ – the Green's function

Definitions A–Z

Adjoint of an operator	Inner product
Adjoint eigenfunctions of the operator	Kronecker delta
Adjoint weighted eigenvalue problem	Levi-Civita symbol
Arc length	Limit definition of the curl
Circulation around a curve	Limit definition of the divergence
Closed surface	Limit definition of the gradient
Conservative vector field	Neumann boundary conditions
Curl in 3D curvilinear coordinates	Non-weighted eigenvalue problem
Curl in Cartesian coordinates	Norm of a vector
Curvilinear coordinates	Normal to a surface
Cylindrical coordinates	Open surface
Dirichlet boundary conditions	Orientable surface
Divergence in 3D curvilinear coordinates	Parametrisation of a curve
Divergence in Cartesian coordinates	Right-handed coordinate system
Divergence theorem	Line integral of a scalar field
Dummy indices	Self-adjoint operator
Eigenfunctions of the operator $\mathcal{L}$	Spherical coordinates
Eigenvalues of an eigenfunction	Stokes' theorem
Free indices	Sturm-Liouville form
Fully self-adjoint operator	Surface integral of a scalar field
Fundamental theorem of line integrals	Surface integral/flux of a vector field
Gradient in Cartesian coordinates	Tangent vector
Gradient in curvilinear coordinates	Vector valued integral
Green's theorem	Line integral of a vector field
Homogeneous boundary conditions	Weighted eigenvalue problem
Inhomogeneous boundary conditions	Weighted inner product

1 Line Integrals

1.1 Curves

Given a curve $C \subset \mathbb{R}^n$, we define a [parametrisation](#) of C ,

$$\mathbf{x}(t) \equiv \sum_{i=1}^n x_i \mathbf{e}_i$$

where $\{\mathbf{e}_i\}_{i=1}^n$ is the standard Cartesian basis. Given a parametrisation, we define the [tangent vector](#) as $\dot{\mathbf{x}}(t)$.

The [line integral of a scalar field](#) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ along a curve C for $t \in [t_0, t_1]$ is given by,

$$\int_C f(\mathbf{x}) ds \equiv \int_{t_0}^{t_1} f(\mathbf{x}) |\dot{\mathbf{x}}| dt$$

and this is independent of the choice of parametrisation. The [arc length](#) s of a curve C from t_0 to t_1 is given by the line integral with $f(\mathbf{x}) \equiv 1$.

The [line integral of a vector field](#) $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ along a curve C for $t \in [t_0, t_1]$ is given by,

$$\int_C \mathbf{F}(\mathbf{x}) \cdot d\mathbf{x} \equiv \int_{t_0}^{t_1} \mathbf{F}(\mathbf{x}) \cdot \dot{\mathbf{x}} dt$$

the [circulation](#) of $\mathbf{F}$ around C is the closed line integral $\oint_C \mathbf{F} \cdot d\mathbf{x}$.

1.2 Curvilinear Coordinates

A coordinate system in $\mathbb{R}^3$ given by $\mathbf{x}(u, v, w)$ is [curvilinear](#) if the coordinate lines (where two of the coordinates are constant) are curved.

The coordinate system $\mathbf{x}(u, v, w)$ is [orthogonal](#) if the tangent vectors $\frac{\partial \mathbf{x}}{\partial u}$, $\frac{\partial \mathbf{x}}{\partial v}$ and $\frac{\partial \mathbf{x}}{\partial w}$ are orthogonal at every point. We then define an orthonormal basis $\{\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w\}$ by

$$\mathbf{e}_u \equiv \frac{1}{h_u} \frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{e}_v \equiv \frac{1}{h_v} \frac{\partial \mathbf{x}}{\partial v}, \quad \mathbf{e}_w \equiv \frac{1}{h_w} \frac{\partial \mathbf{x}}{\partial w}$$

where the scale factors are $h_u \equiv |\frac{\partial \mathbf{x}}{\partial u}|$ and so on.

The following are two key curvilinear coordinate systems.

- [Cylindrical coordinates](#) given by

$$\mathbf{x}(r, \theta, z) \equiv r \cos \theta \mathbf{e}_1 + r \sin \theta \mathbf{e}_2 + z \mathbf{e}_3$$

with $r > 0$, $\theta \in [0, 2\pi]$, $z \in \mathbb{R}$ and scale factors $h_r = 1 = h_z$ and $h_\theta = r$.

- **Spherical coordinates** given by

$$\mathbf{x}(r, \theta, \phi) \equiv r \sin \theta \cos \phi \mathbf{e}_1 + r \sin \theta \sin \phi \mathbf{e}_2 + r \cos \theta \mathbf{e}_3$$

with $r > 0$, $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$ and scale factors $h_r = 1$, $h_\theta = r$ and $h_\phi = r \sin \theta$.

A coordinate system is **right-handed** if

$$\frac{\partial \mathbf{x}}{\partial w} \cdot \left(\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right) > 0$$

2 Surface and Volume Integrals

2.1 Surfaces

Given a 2D surface $S \subset \mathbb{R}^3$ with parametrisation $\mathbf{x}(u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, we define the (unit) **normal** to the surface,

$$\hat{\mathbf{n}} \equiv \frac{\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}}{\left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right|}$$

A surface S together with a consistent choice of normal is an **orientable** surface. Moreover, a surface with no boundary is **open** and a surface with a boundary ∂S is **closed**.

The **surface integral of a scalar field** $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ over a surface S is given by,

$$\int_S f(\mathbf{x}) \, dS \equiv \int_T f(\mathbf{x}(u, v)) \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| \, du \, dv$$

where T is the preimage of S in the (u, v) plane; this is independent of the choice of parametrisation.

The **surface integral/flux of a vector field** $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ over a surface S is given by,

$$\int_S \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S} \equiv \int_T \mathbf{F}(\mathbf{x}(u, v)) \cdot \left(\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right) \, du \, dv$$

where T is the preimage of S in the (u, v) plane; this **depends** on the sign of the (unit) normal.

2.2 Volumes

The **volume integral of a scalar field** $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ over a 3D volume $V \subset \mathbb{R}^3$ is given by,

$$\int_V f(\mathbf{x}) \, dV \equiv \int_T f(\mathbf{x}(u, v, w)) \left| \frac{\partial \mathbf{x}}{\partial u} \cdot \left(\frac{\partial \mathbf{x}}{\partial v} \times \frac{\partial \mathbf{x}}{\partial w} \right) \right| \, du \, dv \, dw$$

where T is the preimage of V in the (u, v, w) space and $\mathbf{x}(u, v, w)$ is the parametrisation of the surface V . Moreover, this is independent of the choice of parametrisation.

For a [vector valued integral](#) of the form,

$$\int_S \mathbf{F} dS \quad \text{or} \quad \int_V \mathbf{F} dV$$

we write out $\mathbf{F}$ in Cartesian components and do three integrals in each component.

3 Differential Operators

3.1 The Gradient

For a scalar field $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, the [gradient in Cartesian coordinates](#) of f is given by,

$$\nabla f \equiv \frac{\partial f}{\partial x} \mathbf{e}_1 + \frac{\partial f}{\partial y} \mathbf{e}_2 + \frac{\partial f}{\partial z} \mathbf{e}_3$$

The [gradient in 3D curvilinear coordinates](#) (u, v, w) is given by,

$$\nabla f \equiv \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w$$

with the scale factors h_u, h_v, h_w and basis vectors $\mathbf{e}_u, \mathbf{e}_v$ and $\mathbf{e}_w$.

We also have a [limit definition of the gradient](#),

$$\nabla f \equiv \lim_{d(S) \rightarrow 0} \frac{1}{|V|} \oint_S f d\mathbf{S}$$

which is useful when proving things but not much in practice.

3.2 The Divergence

For a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the [divergence in Cartesian coordinates](#) of $\mathbf{F}$ is given by,

$$\nabla \cdot \mathbf{F} \equiv \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

where $\mathbf{F} \equiv (F_1, F_2, F_3)$ in the standard basis.

The [divergence in 3D curvilinear coordinates](#) (u, v, w) is given by,

$$\nabla \cdot \mathbf{F} \equiv \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} (F_u h_v h_w) + \frac{\partial}{\partial v} (F_v h_u h_w) + \frac{\partial}{\partial w} (F_w h_u h_v) \right]$$

where $\mathbf{F} \equiv (F_u, F_v, F_w)$ in the (u, v, w) basis.

We also have a [limit definition of the divergence](#),

$$\nabla \cdot \mathbf{F} \equiv \lim_{d(S) \rightarrow 0} \frac{1}{|V|} \oint_S d\mathbf{S} \cdot \mathbf{F}$$

which is again not that useful in practice.

3.3 The Curl

For a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the [curl in Cartesian coordinates](#) of $\mathbf{F}$ is given by,

$$\nabla \times \mathbf{F} \equiv \det \begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{pmatrix}$$

where $\mathbf{F} \equiv (F_1, F_2, F_3)$ in the standard basis.

The [curl in 3D curvilinear coordinates](#) (u, v, w) is given by,

$$\nabla \times \mathbf{F} \equiv \det \begin{pmatrix} h_u \mathbf{e}_u & h_v \mathbf{e}_v & h_w \mathbf{e}_w \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ h_u F_u & h_v F_v & h_w F_w \end{pmatrix}$$

where $\mathbf{F} \equiv (F_u, F_v, F_w)$ in the (u, v, w) basis.

We also have a [limit definition of the curl](#),

$$\nabla \times \mathbf{F} \equiv \lim_{d(S) \rightarrow 0} \frac{1}{|V|} \oint_S d\mathbf{S} \times \mathbf{F}$$

which is yet again not that useful in practice.

3.4 Product Rules

For $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}$ scalar fields and $\mathbf{F}, \mathbf{G} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ we have the [product rules](#),

$$\nabla(fg) \equiv g\nabla f + f\nabla g$$

$$\nabla \cdot (g\mathbf{F}) \equiv (\nabla g) \cdot \mathbf{F} + g(\nabla \cdot \mathbf{F})$$

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) \equiv \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$$

$$\nabla \times (g\mathbf{F}) \equiv g(\nabla \times \mathbf{F}) + (\nabla g) \times \mathbf{F}$$

$$\nabla \times (\mathbf{F} \times \mathbf{G}) \equiv (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F})$$

$$\nabla(\mathbf{F} \cdot \mathbf{G}) \equiv (\mathbf{G} \cdot \nabla)\mathbf{F} + (\mathbf{F} \cdot \nabla)\mathbf{G} + \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F})$$

4 Integral Theorems

4.1 The Fundamental Theorem of Line Integrals

The following is the [fundamental theorem of line integrals](#).

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a scalar field and C is a curve with parametrisation $\mathbf{x}(t)$ for $t \in [t_0, t_1]$, then

$$\int_C \nabla f \cdot d\mathbf{x} \equiv f(\mathbf{x}(t_1)) - f(\mathbf{x}(t_0))$$

We say a vector field $\mathbf{F}$ is **conservative** if there is a scalar field g such that $\mathbf{F} \equiv \nabla g$.

4.2 The Divergence Theorem

The following is the **divergence theorem**.

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a scalar field and V is a volume with **closed** boundary ∂V , then

$$\int_V (\nabla \cdot \mathbf{F}) dV \equiv \oint_{\partial V} \mathbf{F} \cdot d\mathbf{S}$$

It is essential that the boundary is closed. This can be invoked even if the boundary is not connected, for example if $\partial V \equiv \cup_i S_i$.

4.3 Stokes' Theorem and Green's Theorem

The following is **Stokes' theorem**.

If $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a vector field and S is a surface with **closed** boundary ∂S , then

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} \equiv \oint_{\partial S} \mathbf{F} \cdot d\mathbf{x}$$

Similar to the divergence theorem, this can be invoked even if the boundary is not connected, for example if $\partial S \equiv \cup_i C_i$.

If the surface $S \subset \mathbb{R}^2$, then the following is **Green's theorem**.

If $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vector field and C is a closed curve in $\mathbb{R}^2$ oriented anticlockwise, and encloses an area A , then

$$\int_A \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy \equiv \oint_C \mathbf{F} \cdot d\mathbf{x}$$

5 Index Notation

There are two types of indices. **Dummy indices** are indices repeated twice and are summed over - the dimension which is summed over will be clear from context. **Free indices** are indices that appear once and are not summed over. Every index appears no more than twice.

The **Kronecker delta** is given by,

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

The [Levi-Civita symbol](#) is given by,

$$\varepsilon_{ijk} := \begin{cases} 0 & \text{if any two indices are equal} \\ +1 & \text{if } (i, j, k) = (1, 2, 3), (3, 1, 2) \text{ or } (2, 3, 1) \\ -1 & \text{if } (i, j, k) = (1, 3, 2), (2, 1, 3) \text{ or } (3, 2, 1) \end{cases}$$

We have the relation,

$$\varepsilon_{ijk}\varepsilon_{imn} \equiv \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}$$

We have the following.

$$\mathbf{a} \cdot \mathbf{b} \equiv a_i b_i, \quad [\mathbf{a} \times \mathbf{b}]_i \equiv \varepsilon_{ijk} a_j b_k, \quad [\nabla f]_i \equiv \partial_i f, \quad \nabla \cdot \mathbf{F} \equiv \partial_i F_i, \quad [\nabla \times \mathbf{F}]_i \equiv \varepsilon_{ijk} \partial_j F_k.$$

6 Partial Differential Equations

6.1 What is a PDE?

A linear/non-linear PDE is exactly what you think it is.

There are two types of boundary conditions. [Dirichlet boundary conditions](#) restrict the value of a function on the boundary. [Neumann boundary conditions](#) restrict the value of the derivative of a function on the boundary.

[Homogeneous boundary conditions](#) are when all boundary conditions take the value zero. [Inhomogeneous boundary conditions](#) are when any single boundary condition takes a non-zero value.

6.2 Separation of Variables

The following is the method.

1. Use a separation ansatz, for instance $u(x, t) = X(x)T(t)$.
2. Substitute this into the PDE and separate the equation into the addition of distinct modes, for instance $\frac{X''}{X} - \frac{T''}{T} = 0$.
3. Set each mode equal to a constant that satisfies the equation, for instance $\frac{X''}{X} = \lambda_x$, $-\frac{T''}{T} = \lambda_t$ and $\lambda_x + \lambda_t = 0$.
4. Solve the spacial modes and apply the boundary conditions to find the eigenfunctions and eigenvalues.

5. Solve for the temporal modes.
6. Find the general solution by summing over all spacial modes, for instance $u(x, t) = \sum_n X_n(x)T_n(x)$.

7 Boundary Value Problems

7.1 Functional Spaces

Consider an infinite-dimensional vector space of ‘reasonably behaved real functions’ on the domain $a \leq x \leq b$.

We define the following for vectors $u(x), v(x)$ in our vector space.

- The [inner product](#),

$$\langle u, v \rangle \equiv \int_a^b u(x)v(x) \, dx$$

the [weighted inner product](#),

$$\langle u, v \rangle_r \equiv \int_a^b u(x)v(x)r(x) \, dx$$

where $r(x)$ is known as the weight which is real and non-negative on the domain.

- The [norm](#) of a vector $u(x)$ as $\|u(x)\| \equiv \sqrt{\langle u, u \rangle}$.

7.2 Operators and Adjoint

We will consider second order linear operators but the following generalises to an arbitrary order.

For a linear operator $\mathcal{L}$ acting on our space with boundary conditions BC_i , we define the [adjoint](#) of $\mathcal{L}$ as the linear operator $\mathcal{L}^*$ such that

$$\langle \mathcal{L}[u], v \rangle = \langle u, \mathcal{L}^*[v] \rangle + \text{boundary terms}$$

where the boundary terms are evaluated at the end points of the domain. We then force these terms to be zero, which give the boundary conditions of $\mathcal{L}^*$ denoted BC_i^* .

Importantly, every linear operator has associated boundary conditions.

An operator is said to be [self-adjoint](#) if $\mathcal{L} = \mathcal{L}^*$ and [fully self-adjoint](#) if $\mathcal{L} = \mathcal{L}^*$ and further that $\text{BC}_i = \text{BC}_i^*$.

To find the adjoint and associated boundary conditions, (possibly repeated) integration by parts is used along with the boundary conditions BC_i when forcing the boundary terms to vanish.

7.3 Eigenvalue Problems

For a linear operator $\mathcal{L}$ with associated boundary conditions BC_i , the [weighted eigenvalue problem](#) of $\mathcal{L}$ with weight $r(x)$ is given by

$$\mathcal{L}[y_n(x)] = \lambda_n r(x) y_n(x)$$

where if $r(x) \equiv 1$, then this is known as an [non-weighted eigenvalue problem](#). The functions that satisfy this relation $\{y_n\}$ are known as [eigenfunctions of the operator \$\mathcal{L}\$](#) with the [eigenvalues of the eigenfunctions](#) being λ_n .

Related to the linear operator $\mathcal{L}$, the [adjoint weighted eigenvalue problem](#) for the linear operator $\mathcal{L}^*$ is given by

$$\mathcal{L}^*[w_n(x)] = \lambda_n r(x) w_n(x)$$

where the eigenvalues of the adjoint problem are the same as the original problem. The functions $\{w_n\}$ that satisfy this relation are known as [adjoint eigenfunctions of the operator \$\mathcal{L}\$](#) with the eigenvalues of the eigenfunctions being λ_n .

Furthermore, we have the following relating the eigenvalue problem to the adjoint eigenvalue problem.

- Eigenfunctions corresponding to different eigenvalues are orthogonal to the adjoint eigenfunctions with weight $r(x)$, that is

$$\langle y_i, w_j \rangle_r = 0$$

for y_i and w_j having distinct eigenvalues.

In practice, finding the eigenfunctions and eigenvalues of an operator requires solving the differential equation and applying the boundary conditions.

If an operator $\mathcal{L}$ is fully self-adjoint, then the eigenfunctions of the original problem are equal to the eigenfunctions of the adjoint problem, that is $y_n(x) = w_n(x)$ for all n .

7.4 Sturm-Liouville Theory

A linear operator $\mathcal{L}$ is in [Sturm-Liouville form](#) if

$$\mathcal{L}[y] \equiv \frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + q(x)y$$

for some functions $p(x)$ and $q(x)$. The operator is fully self-adjoint under the boundary conditions

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0$$

$$\alpha_3 y(b) + \alpha_4 y'(b) = 0$$

for some constants α_i .

In order for a linear second order operator of the form

$$\mathcal{L}[y] \equiv a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y$$

for some functions $a_i(x)$ to be in Sturm-Liouville form, we can multiply it by a factor $\mu(x)$, and equate coefficients to the Sturm-Liouville form. This leads to the operator $\mu\mathcal{L}$ being in Sturm-Liouville form.

7.5 Solving Boundary Value Problems

7.5.1 Homogeneous boundary conditions

If we have a boundary value problem of the form

$$\mathcal{L}[u(x)] = f(x)$$

with linear homogeneous separated boundary conditions $BC_1[u] = BC_2[u] = 0$, we can solve it using the following.

1. Solve the weighted eigenvalue problem

$$\mathcal{L}[y_n] = \lambda_n r(x) y_n$$

with the boundary conditions $BC_1[y_n] = BC_2[y_n] = 0$ to find the eigenfunctions and eigenvalues.

2. Find the adjoint $\mathcal{L}^*$ with boundary conditions BC_1^* and BC_2^* .
3. Solve the weighted adjoint eigenvalue problem,

$$\mathcal{L}^*[w_n] = \lambda_n r(x) w_n$$

with the boundary conditions $BC_1^*[w_n] = BC_2^*[w_n] = 0$ to find the eigenfunctions.

4. Assume a solution of the form $u(x) = \sum_i c_i y_i(x)$ and take an inner product to find the coefficients c_i ,

$$\begin{aligned} \mathcal{L}[u] = f(x) &\iff \langle \mathcal{L}[u], w_n \rangle = \langle f(x), w_n \rangle \iff \langle u, \mathcal{L}^*[w_n] \rangle = \langle f(x), w_n \rangle \\ &\iff \langle u, \lambda_n r(x) w_n \rangle = \langle f(x), w_n \rangle \iff \sum_i \langle c_i y_i, \lambda_n r(x) w_n \rangle = \langle f(x), w_n \rangle \\ &\iff c_n \lambda_n \langle y_n, r(x) w_n \rangle = \langle f(x), w_n \rangle \end{aligned}$$

5. For $\lambda_n \neq 0$, we have

$$c_n = \frac{\langle f(x), w_n \rangle}{\lambda_n \langle y_n, r(x) w_n \rangle}$$

7.5.2 Inhomogeneous boundary conditions

If instead we have a boundary value problem of the form

$$\mathcal{L}[u(x)] = f(x)$$

with linear boundary conditions $\text{BC}_1[u] = \gamma_1$ and $\text{BC}_2[u] = \gamma_2$, we can solve it breaking the problem into two parts.

1. A homogeneous problem for $u_1(x)$,

$$\mathcal{L}[u_1] = f(x)$$

with $\text{BC}_1[u] = \text{BC}_2[u] = 0$.

2. An inhomogeneous problem for $u_2(x)$,

$$\mathcal{L}[u_2] = 0$$

with $\text{BC}_1[u] = \gamma_1$ and $\text{BC}_2[u] = \gamma_2$.

We then have that $u(x) = u_1(x) + u_2(x)$ is a solution by linearity.

7.6 The Fredholm Alternative

When solving the inhomogeneous boundary value problem

$$\mathcal{L}[u] = f(x)$$

with boundary conditions BC_i , one of the two hold.

1. The problem has a unique solution.

2. The homogeneous adjoint problem

$$\mathcal{L}^*[w] = 0$$

has a non-trivial solution. Moreover,

(a) If $\langle w, f \rangle = 0$ then we have a non-unique solution to the problem.

(b) If $\langle w, f \rangle \neq 0$ then we have no solution to the problem.

8 Green's Method

8.1 The General Method

8.1.1 Homogeneous

We wish to solve the boundary value problem

$$\mathcal{L}[y(x)] = f(x)$$

with $\mathcal{L} = a_n(x)\frac{d^n}{dx^n} + \cdots + a_1(x)\frac{d}{dx} + a_0(x)$ and $n - 1$ **homogeneous** boundary conditions $BC_i[y] = 0$ on $x \in [a, b]$. We have the following method.

1. Let $\mathcal{G}(x, \xi)$ be the Green's function.
2. The Green's function satisfies

$$\mathcal{L}[\mathcal{G}(x, \xi)] = \delta(x - \xi)$$

which we can solve on the divided domain by solving

$$\mathcal{L}[\mathcal{G}(x, \xi)] = 0$$

for $x \in [a, \xi) \cup (\xi, b]$.

3. The $n - 1$ boundary conditions on x , $BC_i[\mathcal{G}(x, \xi)] = 0$.
4. The jump condition on the $n - 1^{th}$ partial derivative

$$\lim_{x \rightarrow \xi^+} \frac{\partial^{n-1} \mathcal{G}(x, \xi)}{\partial x^{n-1}} - \lim_{x \rightarrow \xi^-} \frac{\partial^{n-1} \mathcal{G}(x, \xi)}{\partial x^{n-1}} = \frac{1}{a_n(\xi)}$$

5. The continuity condition on all derivatives lower than $n - 1$ at $x = \xi$,

$$\lim_{x \rightarrow \xi^-} \frac{\partial^j \mathcal{G}(x, \xi)}{\partial x^j} = \lim_{x \rightarrow \xi^+} \frac{\partial^j \mathcal{G}(x, \xi)}{\partial x^j}$$

for $j = 0, 1, \dots, n - 2$

Our final solution is then

$$y(x) = \int_a^b \mathcal{G}(x, \xi) f(\xi) d\xi$$

8.1.2 Inhomogeneous

We wish to solve the boundary value problem

$$\mathcal{L}[y(x)] = f(x)$$

with $\mathcal{L} = a_n(x) \frac{d^n}{dx^n} + \cdots + a_1(x) \frac{d}{dx} + a_0(x)$ and $n - 1$ **inhomogeneous** boundary conditions $\text{BC}_i[y] = \gamma_i$ on $x \in [a, b]$. We have the following method.

1. Solve the homogeneous boundary value problem

$$\mathcal{L}[y_h(x)] = f(x)$$

with boundary conditions $\text{BC}_i[y_h] = 0$ using the method above to find $y_h(x)$.

2. Solve the inhomogeneous boundary value problem

$$\mathcal{L}[y_{ih}(x)] = 0$$

with boundary conditions $\text{BC}_i[y_{ih}] = \gamma_i$ using any valid method.

3. Final answer is then

$$y(x) = y_h(x) + y_{ih}(x)$$

by the principle of superposition.